

COMPUTER ARCHITECTURE

EE-3009



TRENDS IN TECHNOLOGY

- If an instruction set architecture is to prevail, it must be designed to survive rapid changes in computer technology.
- An architect must plan for technology changes that can increase the lifetime of a successful computer.
- To plan for the evolution of a computer, the designer must be aware of rapid changes in implementation technology. Five implementation technologies, which change at a dramatic pace, are critical to modern implementations:

CLASS ACTIVITY

- "If a computer has 5 billion transistors today, how many could it have after 10 years if the transistor count doubles every 2 years?"

Moore's Law

- Moore's Law is the prediction that the number of transistors on a chip will **double** roughly every **two** years, with a minimal increase in cost.
- Moore's Law, created by Gordon Moore before he co-founded Intel, has remained the golden rule for the electronics industry since its 1965 publication.
- Moore's Law only stops when innovation stops, and innovation continues to push forward.



MOORE'S LAW

- The mathematical model:
- If transistor count doubles every T years:

$$N(t) = N_0 \times 2^{t/T}$$

- Where: N_0 = initial transistor count
- $N(t)$ = transistor count after time t
- T = doubling period
- t = elapsed time
- For Moore's Law: $T=2$ years

CLASS ACTIVITY

- 1. A processor currently contains **4 billion transistors**. Assuming Moore's Law, estimate the transistor count after **6 years**.
- 2. A CPU has **10 billion transistors**. Another CPU has **80 billion transistors**.
Assuming Moore's Law, approximately how much time separates the two generations?

EXERCISE

- A chip currently contains **3 billion transistors**. Under a two-year doubling rate, after how many years would it theoretically exceed **100 billion transistors**?

$$100 = 3 \times 2^{t/2}$$

$$33.33 = 2^{t/2}$$

$$\frac{t}{2} = \log_2(33.33)$$

$$t = 2 \log_2(33.33)$$

$$t \approx 10.1 \text{ years}$$



PERFORMANCE PER WATT



Strained Silicon



Intel 90nm



Enhanced Strain



Intel 65nm



Hi-K Metal Gate



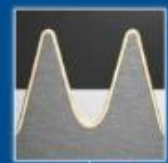
Intel 45nm



Enhanced HKMG



Intel 32nm



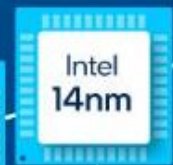
First FinFET



Intel 22nm



Enhanced FinFET



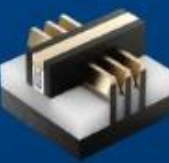
Intel 14nm



Intel 10nm



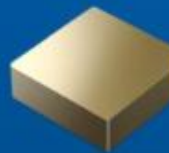
COAG



Enhanced FinFET



Intel 10nm SuperFIN



Super MIM Capacitor



Intel 7

Transistor optimization for performance

Metal stack enhancements



Intel 4



EUV Litho



Intel 3

Denser design libraries

Increased transistor drive current

Reduced via resistance

Increased use of EUV



Intel 20A

RibbonFET + PowerVia



Intel 18A

Ribbon innovation increased performance

Continued metal linewidth reduction



Intel NEXT

Angstrom era

TIME

INTEGRATED CIRCUIT LOGIC TECHNOLOGY

- Historically, transistor density increased by about 35% per year, quadrupling somewhat over four years.
- Increases in die size are less predictable and slower, ranging from 10% to 20% per year.
- The combined effect was a traditional growth rate in transistor count on a chip of about 40%–55% per year, or doubling every 18–24 months.
- This trend is popularly known as **Moore's Law**.

LATENCY AND BANDWIDTH

- Latency = the time you have to wait before an operation starts producing its result/data.
- Latency tells us "How long do I have to wait?"
- Suppose the CPU requests data from RAM and the first byte of data arrives after **100 ns**.

$$\text{Latency} = 100 \text{ ns}$$

- Lower latency = better/faster response.

LATENCY AND BANDWIDTH

- Bandwidth = the amount of data that can be transferred per unit of time.
- Bandwidth tells us "How much data can I transfer per second?"
- It is usually measured in:

MB/s

GB/s

- Higher bandwidth = more data can be transferred per second.

SEMICONDUCTOR **DRAM** (DYNAMIC RANDOM-ACCESS MEMORY)

- This technology is the foundation of main memory.
- The growth of DRAM has slowed dramatically, from quadrupling every three years as in the past.
- The 8-gigabit DRAM was shipping in 2014, but the 16-gigabit DRAM won't reach that state until 2019.

SEMICONDUCTOR FLASH

- electrically erasable programmable read-only memory.
- This nonvolatile semiconductor memory is the standard storage device in PMDs, and its rapidly increasing popularity has fueled its rapid growth rate in capacity.
- In recent years, the capacity per Flash chip increased by about 50%–60% per year, doubling roughly every 2 years.
- Currently, Flash memory is 8–10 times cheaper per bit than DRAM.

MAGNETIC DISK TECHNOLOGY

- Prior to 1990, density increased by about 30% per year, doubling in three years.
- It rose to 60% per year thereafter, and increased to 100% per year in 1996.
- Between 2004 and 2011, it dropped back to about 40% per year, or doubled every two years.
- Recently, disk improvement has slowed to less than 5% per year.
- This technology is central to server- and warehouse-scale storage.

NETWORK TECHNOLOGY

The Evolution of 5G



1980s

1G

Analog
Telecommunications



1990s

2G

Text
Messaging



2000s

3G

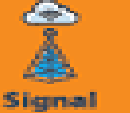
Mobile and Wireless
Internet Connection



2010s

4G

Cloud, IP and Mobile
Broadband



2020s

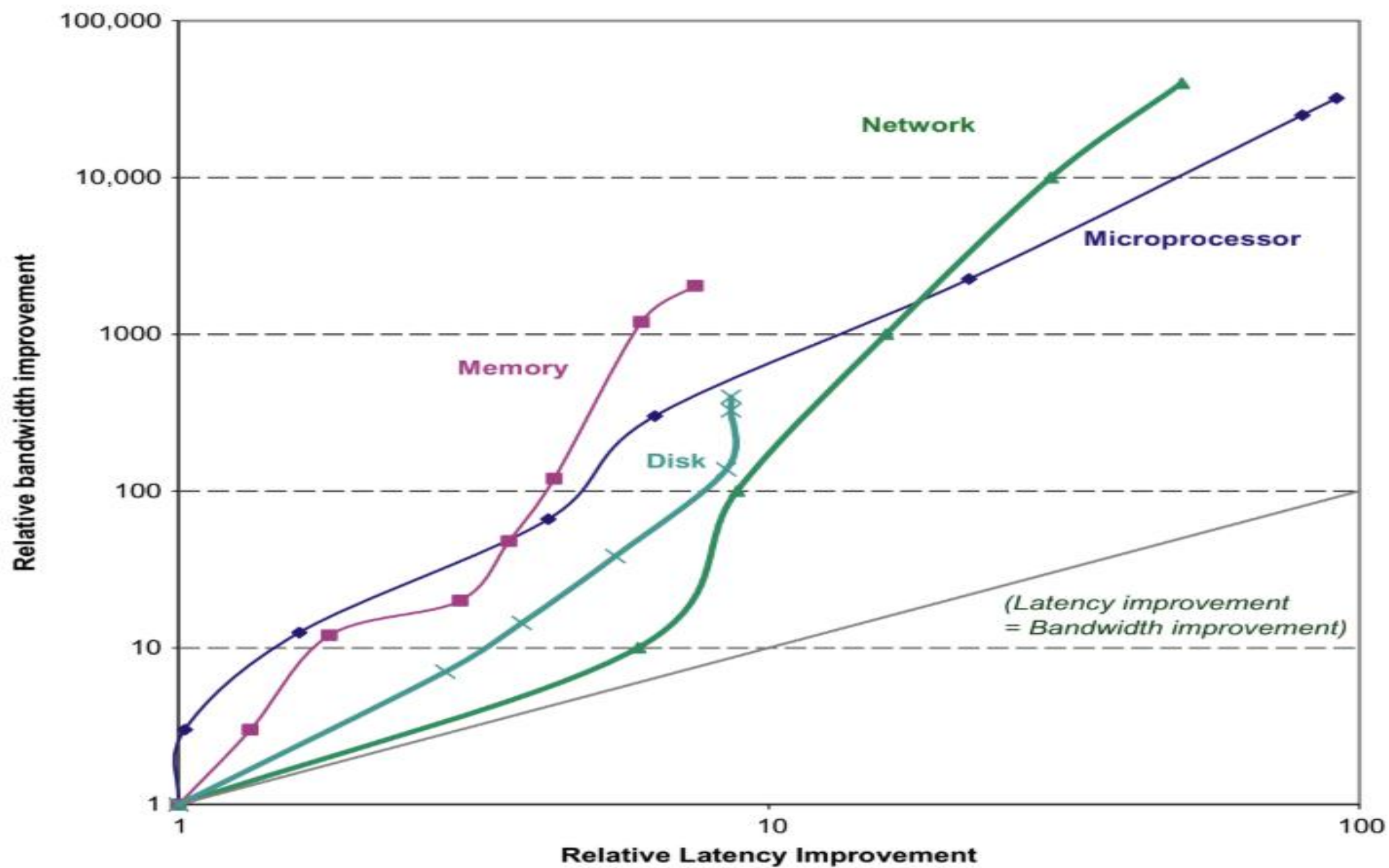
5G

**FASTER
EVERYTHING!**

Unlimited Data Capacity

PERFORMANCE TRENDS: BANDWIDTH OVER LATENCY

- Bandwidth or throughput is the total amount of work done in a given time, such as megabytes per second for a disk transfer.
- In contrast, latency or response time is the time between the start and the completion of an event, such as milliseconds for a disk access.
- A simple rule of thumb is that bandwidth grows by at least the square of the improvement in latency. Computer designers should plan accordingly.



MAXIMUM POWER A PROCESSOR EVER REQUIRES?

A processor does not consume a constant amount of power.

For example:

When the CPU is idle → low power

When running normal programs → moderate power

When performing heavy calculations → high power

During short bursts of activity → it may demand very high peak power

$$P = V \times I$$

MAXIMUM POWER A PROCESSOR EVER REQUIRES?

Modern processors can dynamically adjust:

- Voltage
- Clock frequency
- Power consumption

CPU State	Frequency	Voltage	Performance	Power
Low-power	2.0 GHz	0.8 V	Low	Low
Normal	3.5 GHz	1.0 V	Medium/High	Medium
Boost	5.0 GHz	1.2 V	Very High	High

WHAT IS SUSTAINED POWER CONSUMPTION?

The amount of power a processor is expected to dissipate continuously during a sustained workload.

- TDP = Thermal Design Power
- TDP is primarily a thermal-design target.

For example, suppose a processor has: $TDP = 100W$

This means the computer's cooling system should generally be capable of handling approximately 100 W of sustained thermal load under the relevant design conditions.



Effect of Lower Frequency (Thermal Throttling)



Lower Frequency

CPU clock rate is reduced



Lower Switching Activity

Fewer transitions per second



Lower Power

Power consumption decreases



Less Heat Generation

Less electrical power converted to heat



Temperature Decreases / Stabilizes

Temperature drops or stays within safe limits

This is commonly called
thermal throttling.



1. CPU becomes dangerously hot

Temperature approaches or exceeds safe operating limit.



2. Frequency reduction is attempted

CPU lowers clock rate (thermal throttling) to reduce power and heat.



3. Temperature still continues increasing

Despite frequency reduction, temperature keeps rising.



4. Emergency thermal protection activates

Hardware safety mechanism is triggered to prevent damage.



5. CPU shuts down

Processor powers down immediately to prevent permanent damage.



This is essentially the processor saying:
"If slowing down isn't enough to protect me, I will shut myself down."

WHAT IS SUSTAINED POWER CONSUMPTION?

Cooling system and TDP

- a cooling system is usually designed to match or exceed TDP.

Suppose:

- $TDP = 100W$
- A cooling solution might be designed to handle: 120W or more, depending on the system design.

ENERGY AND ENERGY EFFICIENCY

The key idea is that power and energy are not the same thing, and when comparing processors for a particular task, energy is usually the better metric.

$$Energy = Power \times Time$$

Question : Which Processor is Energy Efficient?

	Processor A	Processor B
Average power	120 W	100 W
Execution time	7 s	10 s

ENERGY AND ENERGY EFFICIENCY

A processor that consumes more power can still be more energy-efficient if it completes the task significantly faster.

$$\text{Energy} = \text{Average Power} \times \text{Execution Time}$$

ENERGY AND POWER WITHIN A MICROPROCESSOR

- For CMOS chips, the traditional primary energy consumption has been in switching transistors, also called dynamic energy. The energy required per transistor is proportional to the product of the capacitive load driven by the transistor and the square of the voltage:

$$\text{Energy}_{\text{dynamic}} \propto \text{Capacitive load} \times \text{Voltage}^2$$

- This equation is the energy of pulse of the logic transition of 0->1->0 or 1->0->1. The energy of a single transition (0->1 or 1->0) is then

$$\text{Energy}_{\text{dynamic}} \propto 1/2 \times \text{Capacitive load} \times \text{Voltage}^2$$

ENERGY AND POWER WITHIN A MICROPROCESSOR

- The power required per transistor is just the product of the energy of a transition multiplied by the frequency of transitions:

$$\text{Power}_{\text{dynamic}} \propto 1/2 \times \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency switched}$$

- For a fixed task, slowing clock rate reduces power, but not energy.
- Clearly, dynamic power and energy are greatly reduced by lowering the voltage, so voltages have dropped from 5 V to just under 1 V in 20 years. The capacitive load is a function of the number of transistors connected to an output and the technology, which determines the capacitance of the wires and the transistors.

EXAMPLE

- Some microprocessors today are designed to have adjustable voltage, so a 15% reduction in voltage may result in a 15% reduction in frequency. What would be the impact on dynamic energy and on dynamic power?

$$\frac{\text{Energy}_{\text{new}}}{\text{Energy}_{\text{old}}} = \frac{(\text{Voltage} \times 0.85)^2}{\text{Voltage}^2} = 0.85^2 = 0.72$$

$$\frac{\text{Power}_{\text{new}}}{\text{Power}_{\text{old}}} = 0.72 \times \frac{(\text{Frequency switched} \times 0.85)}{\text{Frequency switched}} = 0.61$$

EXERCISE

A processor designer wants to reduce dynamic power by 50%. The clock frequency will be reduced by 20%.

By what percentage should the voltage be reduced?

POWER

- Power consumption increases as processor complexity increases
- Number of transistors increases
- Switching frequency increases
- Early microprocessors consumed about 1W
- 80386 microprocessors consumed about 2W
- 3.3GHz Intel Core i7 consumes about 130W
- Must be dissipated from a chip that is about 1.5cm × 1.5cm

MANAGING POWER FOR FURTHER EXPANSION

- Voltage cannot be reduced further .
- Power per chip cannot be increased because the air cooling limit has been reached.
- Therefore, clock frequency growth has slowed .
- Heat dissipation is now the major constraint on using transistors.

ENERGY EFFICIENCY STRATEGIES

- Modern microprocessors offer many techniques to try to improve energy efficiency despite flat clock rates and constant supply voltages:
- 1. ***Do nothing well.*** Most microprocessors today turn off the clock of inactive modules to save energy and dynamic power. For example, if no floating-point instructions are executing, the clock of the floating-point unit is disabled. If some cores are idle, their clocks are stopped.

ENERGY EFFICIENCY STRATEGIES

- 2. **Dynamic voltage and frequency scaling** (DVFS) is the adjustment of power and speed settings on a computing device's various processors, controller chips and peripheral devices to optimize resource allotment for tasks and maximize power saving when those resources are not needed.
- If the current task is computationally intensive, meaning it requires the processor to consume more energy and run at a higher frequency, DVFS will ensure that the voltage and frequency increase to match the requirements. However, if the processor needs to run a low-priority or less computationally intensive task where high speed and throughput are not required, it will adjust accordingly.

ENERGY EFFICIENCY STRATEGIES

- *Design for the typical case.*
- PMDs and laptops are often idle
- Use low power mode DRAM to save energy
- Spin disk at lower rate
- PCs use emergency slowdown if program execution causes overheating

ENERGY EFFICIENCY STRATEGIES

- **Overclocking.**
- Intel started offering Turbo mode in 2008, where the chip decides that it is safe to run at a higher clock rate for a short time, possibly on just a few cores, until temperature starts to rise. For example, the 3.3 GHz Core i7 can run in short bursts for 3.6 GHz. Indeed, the highest-performing microprocessors each year since 2008 have all offered temporary overclocking of about 10% over the nominal clock rate.
- For single-threaded code, these microprocessors can turn off all cores but one and run it faster. Note that, although the operating system can turn off Turbo mode, there is no notification once it is enabled, so the programmers may be surprised to see their programs vary in performance because of room temperature.

ENERGY EFFICIENCY STRATEGIES

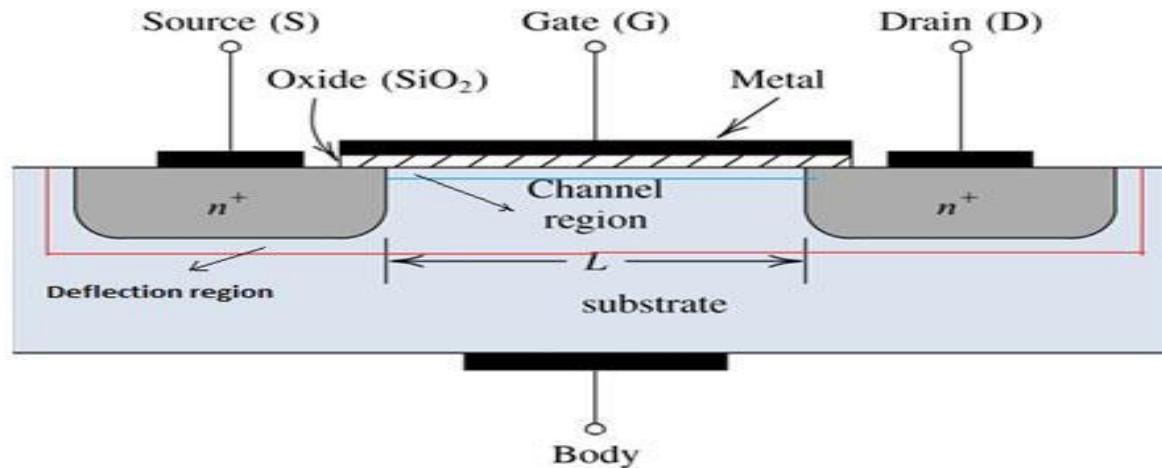
- ***Power gating***
- $\text{Power}_{\text{static}} \propto \text{Current}_{\text{static}} \times \text{Voltage}$
- Current flows in transistors even when idle: leakage current
- Leakage ranges from 25% to 50% of total power
- Power Gating turns off power to inactive modules

ENERGY EFFICIENCY STRATEGIES

- ***Race-to-halt***
- Processor is only part of system cost
- Use faster, less energy-efficient processor to allow the rest of the system to halt

DEPENDABILITY

- Historically, integrated circuits were one of the most reliable components of a computer. Although their pins may be vulnerable, and faults may occur over communication channels, the failure rate inside the chip was very low.

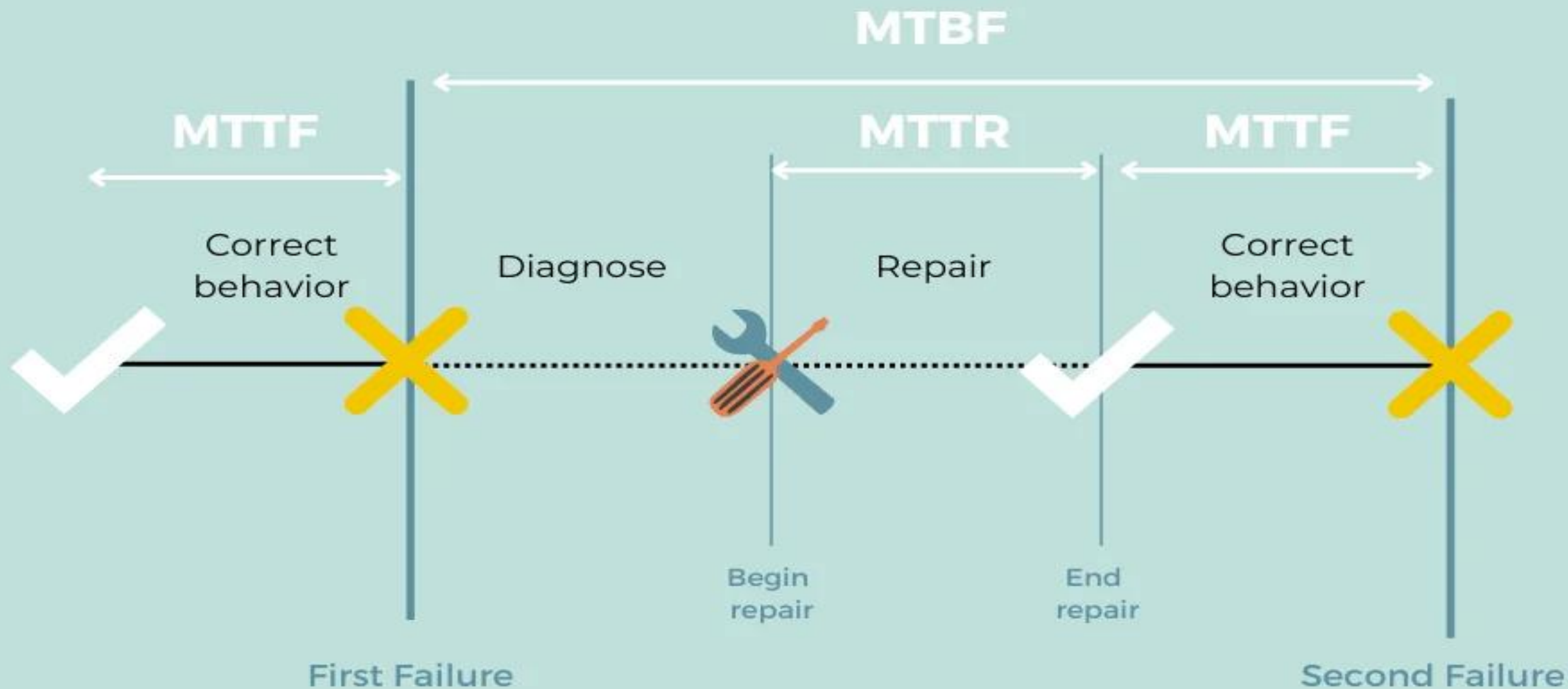


DEPENDABILITY

- Infrastructure providers started offering service level agreements (SLAs) or service level objectives (SLOs) to guarantee that their networking or power service would be dependable. For example, they would pay the customer a penalty if they did not meet an agreement of some hours per month. Thus an SLA could be used to decide whether the system was up or down.
- Systems alternate between two states of service with respect to an SLA:
 1. Service accomplishment, where the service is delivered as specified.
 2. Service interruption, where the delivered service is different from the SLA.

DEPENDABILITY

- Transitions between these two states are caused by failures (from state 1 to state 2) or restorations (2 to 1). Quantifying these transitions leads to the two main measures of dependability:
- Module reliability is a measure of the continuous service accomplishment (or, equivalently, of the time to failure) from a reference initial instant. Therefore the mean time to failure (MTTF) is a reliability measure.



CLASS ACTIVITY

A data center is considering two processor modules:

	Module A	Module B
MTTF	1000 h	1200 h
MTTR	50 h	100 h

Which module has better availability?

EXERCISE

A computer system contains a critical memory module. It fails on average every 1500 hours of operation and takes 6 hours to repair. After a redesign, its MTTF increases to 1800 hours, but because the new design is more complicated, MTTR increases to 12 hours.

Calculate the original MTBF.

Calculate original availability.

Calculate redesigned MTBF.

Calculate redesigned availability.

Did the redesign improve availability?

FAILURE RATE

In Computer Architecture, failure rate tells us **how frequently a component is expected to fail.**

The simplest relationship with MTTF is:

$$\lambda = 1/\text{MTTF}$$

The reciprocal of MTTF is a rate of failures, generally reported as failures per billion hours of operation, or FIT (for failures in time).

EXAMPLE

- Assume a disk subsystem with the following components and MTTF:
- 10 disks, each rated at 1,000,000-hour MTTF
- 1 ATA controller, 500,000-hour MTTF
- 1 power supply, 200,000-hour MTTF
- 1 fan, 200,000-hour MTTF
- 1 ATA cable, 1,000,000-hour MTTF

Compute the MTTF of the system as a whole.

SOLUTION

Using the simplifying assumptions that the lifetimes are exponentially distributed and that failures are independent, compute the MTTF of the system as a whole.

The sum of the failure rates is

$$\begin{aligned}\text{Failure rate}_{\text{system}} &= 10 \times \frac{1}{1,000,000} + \frac{1}{500,000} + \frac{1}{200,000} + \frac{1}{200,000} + \frac{1}{1,000,000} \\ &= \frac{10 + 2 + 5 + 5 + 1}{1,000,000 \text{ hours}} = \frac{23}{1,000,000} = \frac{23,000}{1,000,000,000 \text{ hours}}\end{aligned}$$

or 23,000 FIT. The MTTF for the system is just the inverse of the failure rate

$$\text{MTTF}_{\text{system}} = \frac{1}{\text{Failure rate}_{\text{system}}} = \frac{1,000,000,000 \text{ hours}}{23,000} = 43,500 \text{ hours}$$

or just under 5 years.